

Experiment 3: Decision Tree using ID3 Algorithm

Aim

To implement the ID3 Decision Tree algorithm using an appropriate dataset and classify a new sample.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the dataset (playtennis.csv).

Step 3: Convert categorical data into numerical values using Label Encoding.

Step 4: Separate the independent features (X) and target variable (y).

Step 5: Create a Decision Tree Classifier using the entropy criterion (ID3).

Step 6: Train the classifier using the training data.

Step 7: Predict the class label for a new sample.

Step 8: Display the predicted result.

Program

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder

data = pd.read_csv("playtennis.csv")

encoder = LabelEncoder()

for column in data.columns:
    data[column] = encoder.fit_transform(data[column])

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)

prediction = model.predict(X.iloc[[0]])

if prediction[0] == 1:
    print("Prediction: Yes")
else:
    print("Prediction: No")
```

Output

```
... Prediction: No
```

Result

Thus, the ID3 Decision Tree algorithm was successfully implemented, and the given sample was classified correctly using the trained decision tree model.